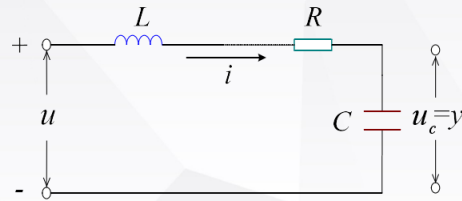


现代控制理论第一次作业：

作业1:求图示RLC回路的状态空间模型



若 $R=3\Omega, L=1H, C=0.5F$, 计算系统的传递函数.

解：

$$i = C \frac{du_c}{dt}, \quad u = u_c + iR + L \frac{di}{dt},$$

$$\text{令 } x_1 = i, x_2 = u_c,$$

$$\text{可得 } C\dot{x}_2 = x_1, \quad u = x_2 + Rx_1 + L\dot{x}_1$$

整理得：

$$\begin{aligned}\dot{x}_1 &= -\frac{R}{L}x_1 - \frac{1}{L}x_2 + \frac{1}{L}u \\ \dot{x}_2 &= \frac{1}{C}x_1 \\ y &= x_2\end{aligned}$$

$$\text{即 } A = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, \quad C = [0 \quad 1], \quad D = 0$$

取 $R = 3\Omega, L = 1H, C = 0.5F$, 代入可得传递函数为：

$$g(s) = C(sI - A)^{-1}B = \frac{2}{s^2 + 3s + 2}$$